Born approximation 1

Derive the Born approximation

$$\psi(r,\theta,\phi)=\frac{e^{ikr}}{r}f(\theta,\phi)$$

where

$$k=\frac{\sqrt{2mE}}{\hbar}$$

and $f(\theta,\phi)$ is the scattering amplitude

$$f(\theta,\phi)=-\frac{m}{2\pi\hbar^2}\int\exp\left(\frac{i\mathbf{p}\cdot\mathbf{x}}{\hbar}\right)V(\mathbf{x})\,d\mathbf{x}$$

Vector $\mathbf{p}$ is momentum transfer such that

$$\mathbf{p}=\mathbf{p}_i-\mathbf{p}_f$$

and

$$p^2=|\mathbf{p}|^2=4mE(1-\cos\theta)$$

§1

Start with the Schrodinger equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x})+V(\mathbf{x})\psi(\mathbf{x})=E\psi(\mathbf{x})$$

Multiply through by $-2m/\hbar^2$ to obtain

$$\nabla^2\psi(\mathbf{x})-\frac{2m}{\hbar^2}V(\mathbf{x})\psi(\mathbf{x})=-\frac{2mE}{\hbar^2}\psi(\mathbf{x})$$

Rewrite as

$$\left(\nabla^2+k^2\right)\psi(\mathbf{x})=\frac{2m}{\hbar^2}V(\mathbf{x})\psi(\mathbf{x})$$

where

$$k=\frac{\sqrt{2mE}}{\hbar}$$

By the sifting property of $\delta(\mathbf{x}-\mathbf{y})$ we can make the right-hand side an integral over $\mathbf{y}$.

$$\left(\nabla^2+k^2\right)\psi(\mathbf{x})=\frac{2m}{\hbar^2}\int\delta(\mathbf{x}-\mathbf{y})V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

Let $G(\mathbf{x})$ be a Green's function such that

$$\left(\nabla^2+k^2\right)G(\mathbf{x})=\delta(\mathbf{x})$$

Then by substitution

$$\left(\nabla^2 + k^2\right)\psi(\mathbf{x}) = \frac{2m}{\hbar^2}\int \left(\nabla^2 + k^2\right) G(\mathbf{x}-\mathbf{y})V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

By linearity the integral can be factored as

$$\left(\nabla^2 + k^2\right)\psi(\mathbf{x}) = \left(\nabla^2 + k^2\right)\frac{2m}{\hbar^2}\int G(\mathbf{x}-\mathbf{y})V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

Hence

$$\psi(\mathbf{x}) = \frac{2m}{\hbar^2}\int G(\mathbf{x}-\mathbf{y})V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

The solution for $G(\mathbf{x})$ is

$$G(\mathbf{x}) = -\frac{\exp(ik|\mathbf{x}|)}{4\pi|\mathbf{x}|}$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2}\int \frac{\exp(ik|\mathbf{x}-\mathbf{y}|)}{|\mathbf{x}-\mathbf{y}|}V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

For scattering experiments we can use the approximation

$$\frac{\exp(ik|\mathbf{x}-\mathbf{y}|)}{|\mathbf{x}-\mathbf{y}|}V(\mathbf{y}) \approx \frac{1}{|\mathbf{x}|}\exp\left(ik|\mathbf{x}| - \frac{ik\mathbf{x}\cdot\mathbf{y}}{|\mathbf{x}|}\right)V(\mathbf{y})$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2}\frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|}\int \exp\left(-\frac{ik\mathbf{x}\cdot\mathbf{y}}{|\mathbf{x}|}\right)V(\mathbf{y})\psi(\mathbf{y})\,d\mathbf{y}$$

For a first-order approximation we substitute a plane wave for $\psi(\mathbf{y})$.

$$\psi(\mathbf{y}) = \exp\left(\frac{ik\mathbf{x}'\cdot\mathbf{y}}{|\mathbf{x}'|}\right)$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2}\frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|}\int \exp\left(\frac{ik\mathbf{x}'\cdot\mathbf{y}}{|\mathbf{x}'|} - \frac{ik\mathbf{x}\cdot\mathbf{y}}{|\mathbf{x}|}\right)V(\mathbf{y})\,d\mathbf{y}$$

Let $\mathbf{p}$ be momentum transfer

$$\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f = \hbar k\left(\frac{\mathbf{x}'}{|\mathbf{x}'|} - \frac{\mathbf{x}}{|\mathbf{x}|}\right)$$

Then by substitution

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2}\frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|}\int \exp\left(\frac{i\mathbf{p}\cdot\mathbf{y}}{\hbar}\right)V(\mathbf{y})\,d\mathbf{y}$$

In polar coordinates

$$\psi(r,\theta,\phi) = -\frac{m}{2\pi\hbar^2}\frac{\exp(ikr)}{r}\int \exp\left(\frac{i\mathbf{p}\cdot\mathbf{y}}{\hbar}\right)V(\mathbf{y})\,d\mathbf{y}$$

§2

Cross sections are completely determined by the scattering amplitude.

$$\frac{d\sigma}{d\Omega} = |\psi(r,\theta,\phi)|^2 \, r^2 = |f(\theta,\phi)|^2$$

For example, for Coulomb potentials we have

$$f(\theta,\phi) = \frac{Z\alpha\hbar c}{2E(1-\cos\theta)}$$

Hence the Rutherford cross section

$$\frac{d\sigma}{d\Omega} = |f(\theta,\phi)|^2 = \left[\frac{Z\alpha\hbar c}{2E(1-\cos\theta)}\right]^2$$

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